Assignment: Ridge Regression using Gradient Descent

1 Objective

The goal of this assignment is to implement **Ridge Regression using Gradient Descent** and analyze its behavior.

You are provided with a template file `ridge.py`. Your task is to complete the missing parts.

2 Background

Ridge Regression minimizes the following loss function:

$$L(w) = \frac{1}{N}\sum_{i=1}^{N}(x_i^T w - y_i)^2 + \lambda||w||^2$$

where:

- $X \in R^{N\times d}$ is the feature matrix
- $y \in R^N$ is the target vector
- $w \in R^d$ is the weight vector
- λ is the regularization parameter

3 Tasks

3.1 Task 1: Compute MSE Loss

Implement the function:

`compute_mse_loss()`

This should compute:

$$\frac{1}{N}\sum(Xw - y)^2$$

3.2 Task 2: Compute Gradient

Implement:

`linear_gradient()`

Gradient of Ridge loss:

$$\nabla L(w) = \frac{2}{N}X^T(Xw - y) + 2\lambda w$$

3.3 Task 3: Gradient Descent

Complete the `fit()` function:

- Initialize weights to zero
- Iteratively update:
$$w = w - \eta \cdot \nabla L(w)$$
- Store loss at each iteration
- Stop when:
  - Maximum iterations reached OR
  - Change in weights is below tolerance

3.4 Task 4: Prediction

Implement:

`predict()`

Return:

$$\hat{y} = Xw$$

4 Files Provided

- `ridge.py` — Implementation file
- `main.py` — For testing
- `data.csv` — Dataset

5 Evaluation

Your implementation will be evaluated based on:

- Correctness of gradient
- Decreasing loss over iterations
- Proper use of regularization